

SP!ED 2025
Team - 2 (One Asia)

The Final Presentation

Real-Time AI Smart Crosswalk(Light)

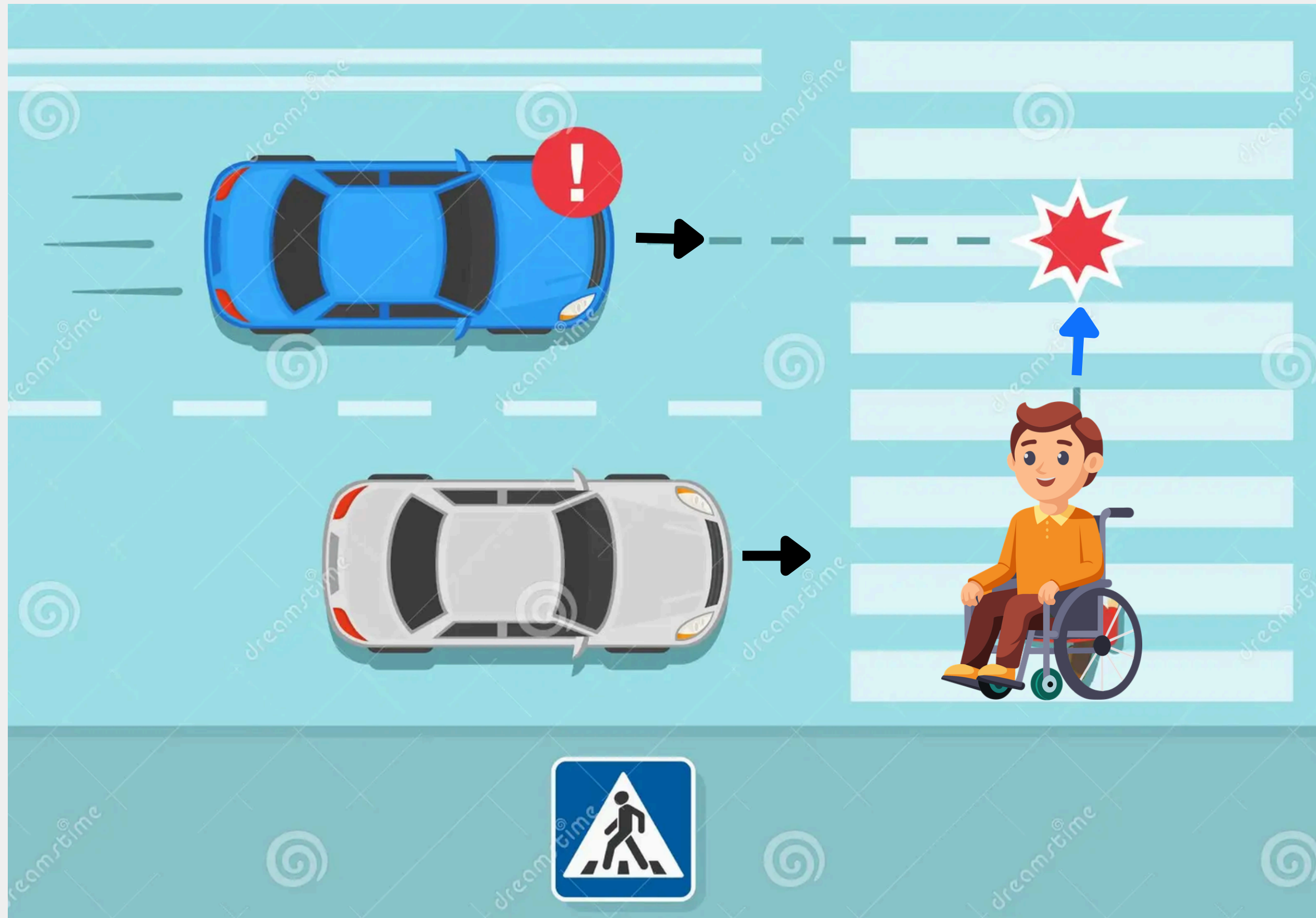


Background

Higher Traffic Risk for Persons with Disabilities

- Mortality $\approx 3.1\times$ higher (vs non-disabled).
- Crashes $\approx 10\times$ higher — 7.7% vs 0.7%.



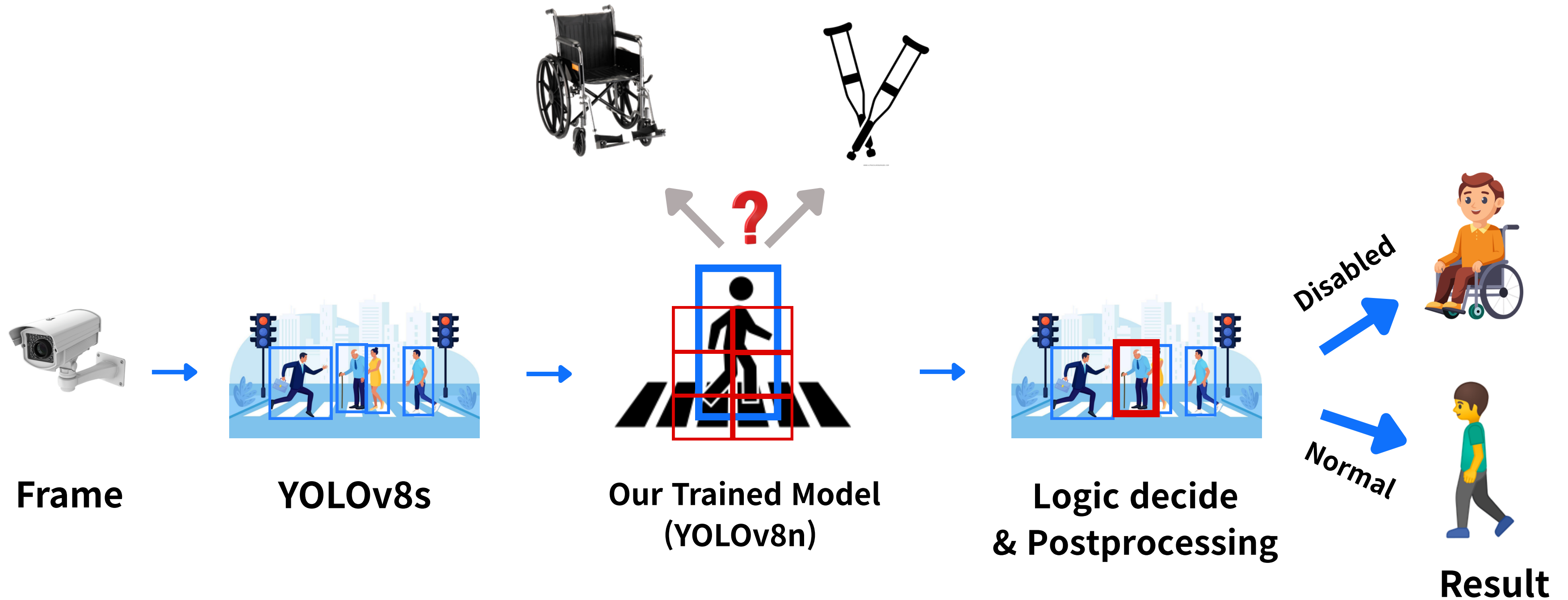


Most frequent crash situations involving persons with disabilities.

Common Accident Scenarios



Dataset: Images 5000+



Frame → YOLO(person detect) → Track(ID) → Pose Estimation → Yolo Model (ours) → Overlay

See Vulnerable Pedestrians—in Rain or at Night

Weather-augmented YOLO + pose estimation for reliable wheelchair/cane detection in low-light and rain/fog.



Frame data



Data Augmentation
Exposure, Hue, Brightness
(+Pose Estimation)

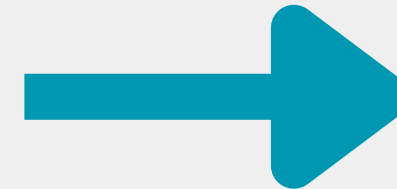


Real Results
 JSU, China

Additional Features



China
Amap

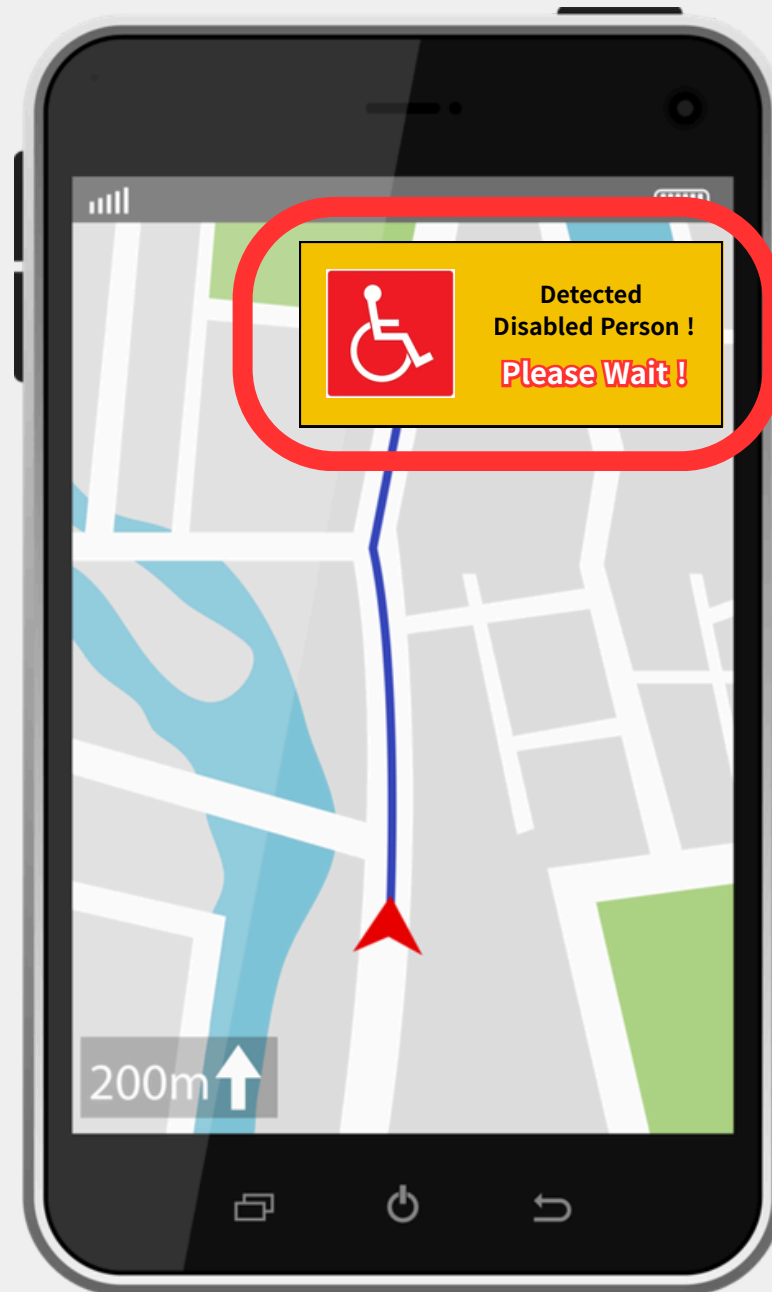


Ours → Integrate with Apps

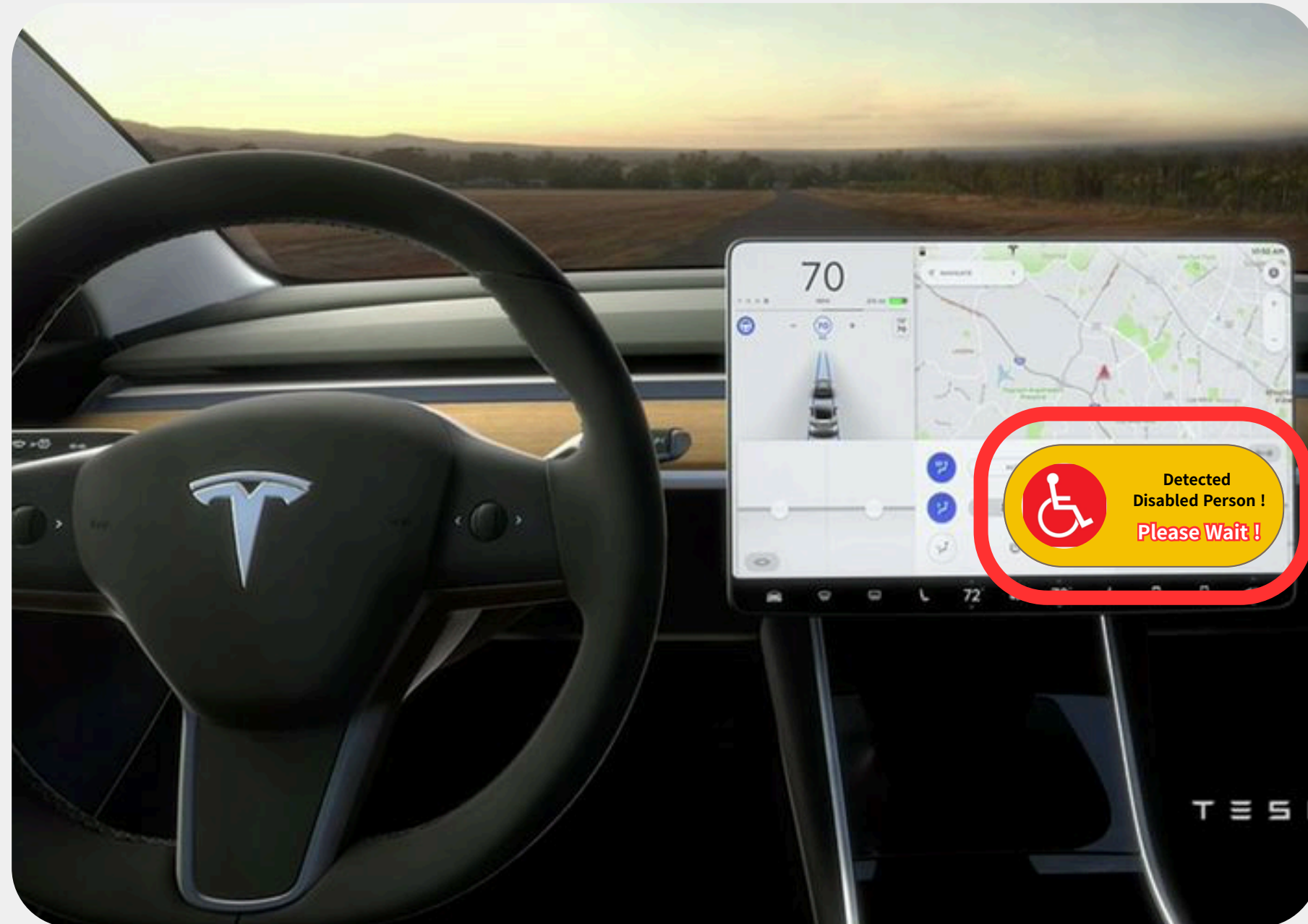
When using map navigation in China, direction, speed, speed limit and time are generally displayed. After using our system, we will give drivers some relevant tips, as shown in the picture on the right. At the same time, some of our information will be displayed on the mobile phone and vehicle navigation. This can be used in google map, Amap, kakaoMap, NaverMap.

Integrate with

We will integrate with Google Map, Amap, KakaoMap, NaverMap

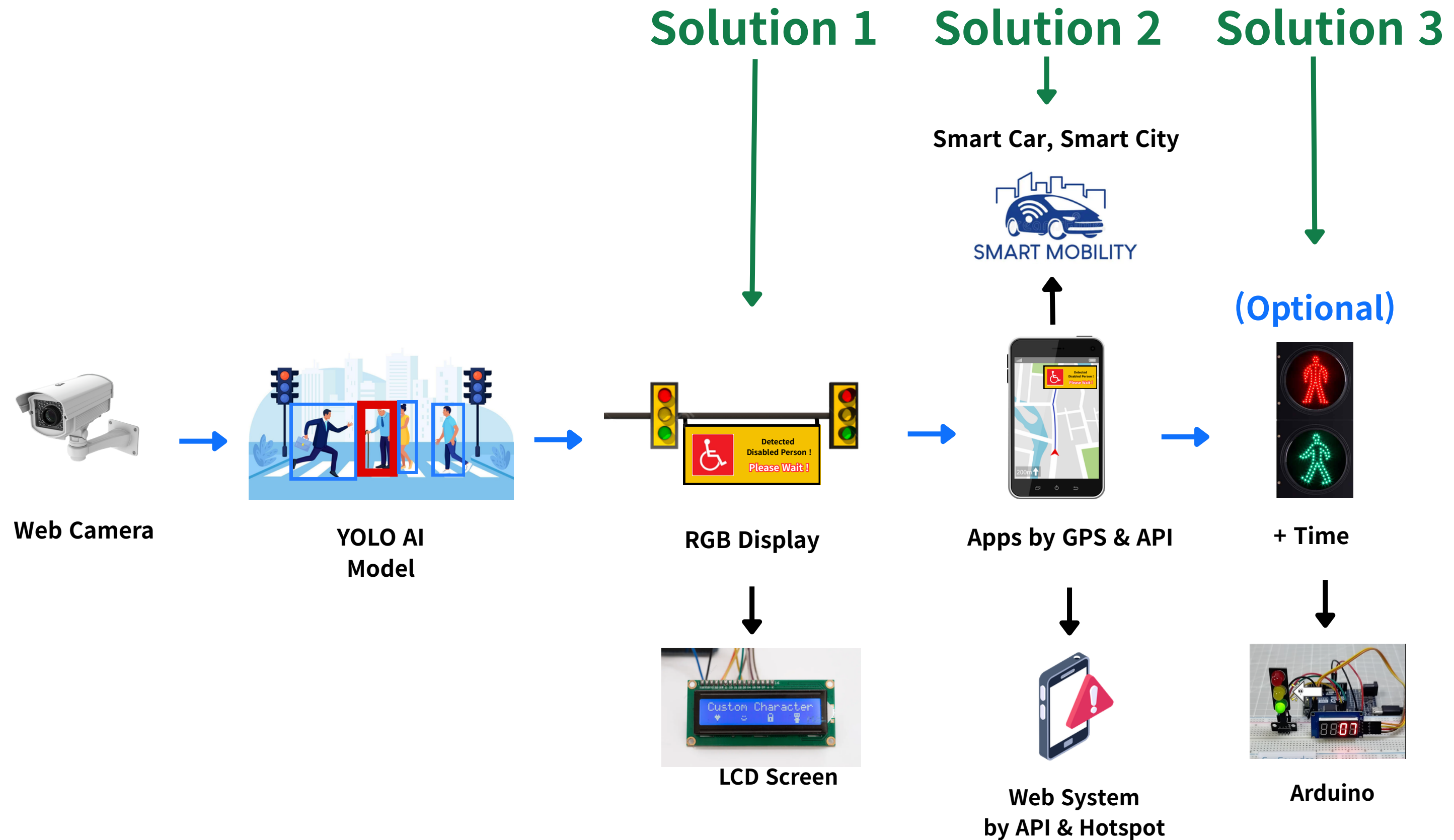


Navigation Apps for **Mobile**



Navigation Apps for **Cars**

System architecture

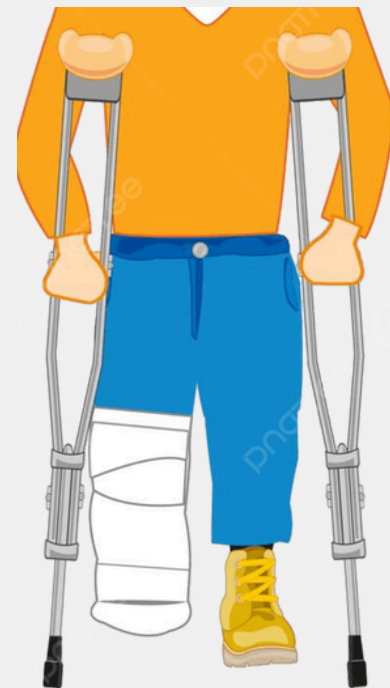


How system works?



Privacy & Data Use

- No face recognition **X**
- Only body(device) images may be stored (no identity)
- Data used only for AI training & safety improvement



Device picture



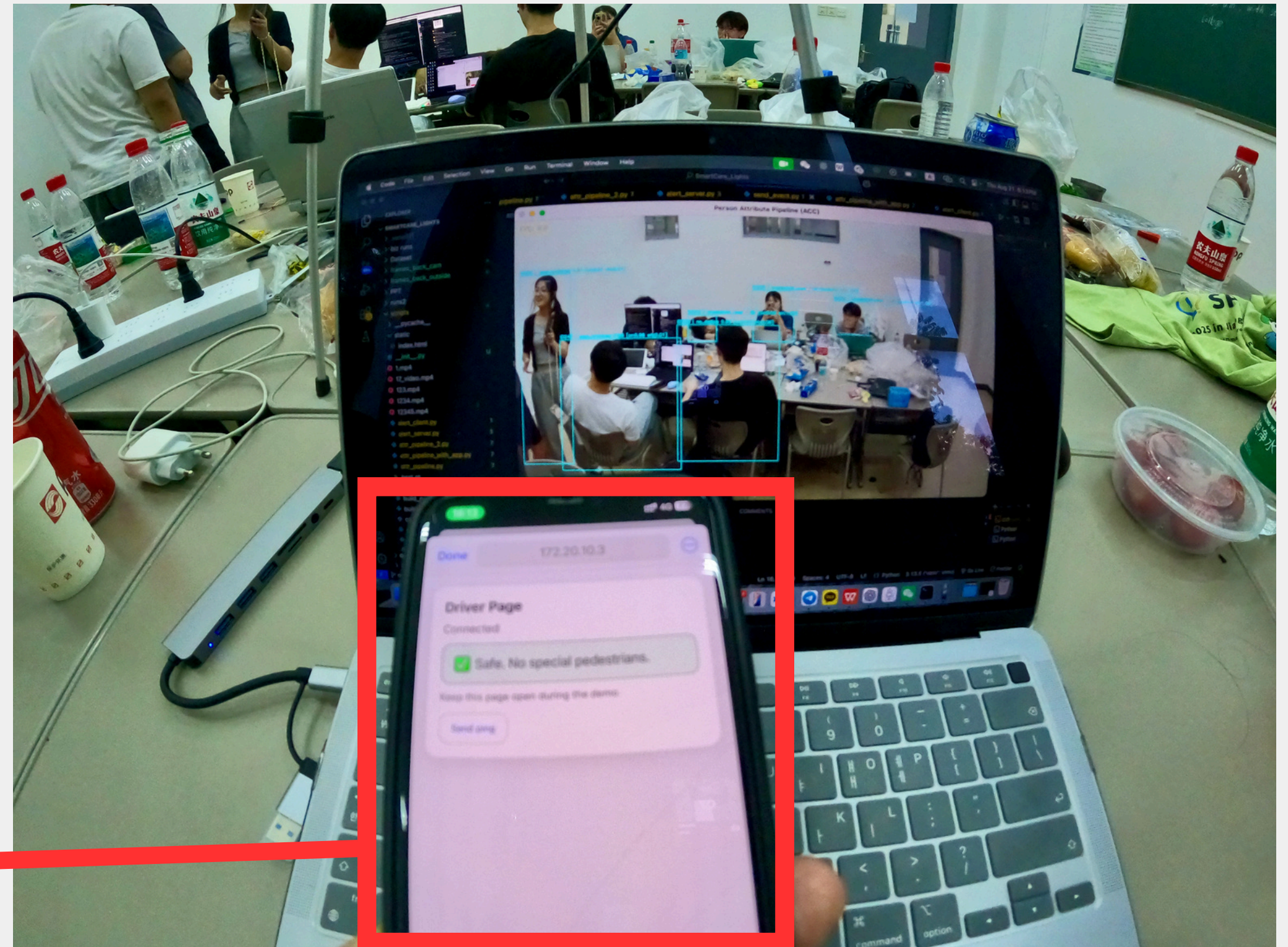
Real Data

Notification



Warning distance depends on speed limit
~100m (50 km/h), ~150m (60–70 km/h)

Testing with Mobile
FastAPI & Hotspot





Macbook Air M1

EXPLORER

SMARTCARE_LIGHTS

runs2

scripts

__pycache__

static

__init__.py

1.mp4

17_video.mp4

123.mp4

1234.mp4

12345.mp4

alert_client.py1

alert_server.py3

attr_pipeline_2.py7, M

attr_pipeline_with_app.py7, M

attr_pipeline.py7

best.pt

best2.pt

best3.ptU

build_no_device_from_uncropped...

build_positive_person_crops.py

calc_class.py

change_class.py

detect_assistive.py2, M

alert_server.py3

send_event.py1

best3.pt

scripts > attr_pipeline_with_app.py > ...
38 # =====
39 # ===== CONFIG =====
40 # =====
41 VIDEO_SOURCE = 0 # 0=webcam yoki video yo'li
42
43 # DETECTOR (person)
44 PERSON_CONF = 0.45
45 MIN_PERSON_AREA = 40 * 40
46 DETECT_IMGSZ = 640
47 USE_HALF = True
48
49 # CLASSIFIER (MNv3) – bazaviy thresholdlar (scale-aware down/up qiladi)
50 CLS_IMSIZE = 224
51 TH_CRUTCHES_BASE = 0.50
52 TH_WC_BASE = 1 # avvalgidan balandroq – WC FPs kamayadi
53
54 # DEVICE-GATING
55 USE_DEVICE_GATING = True
56 DEVICE_CONF = 0.40
57 HARD_GATE_GLOBAL = False # global default; scale-aware qismda WC uchun majb
58 SOFT_GATE_ADD = 0.12
59
60 # TRACKING + TEMPORAL (histerezis)
61 MATCH_MAX_DIST = 160
62 TRACK_MAX_AGE = 24
63 TEMP_WIN = 24 # oynaning uzunligi

It was run **without** a GPU.

Expected Effect

- **Society:** Safer crossing for elderly & disabled, fewer accidents.
- **Drivers:** Display alerts → more awareness, smoother driving.
- **Business:** Low-cost, scalable. Should be required at hospitals, schools, elderly homes → strong market.

Real System (Average hardware cost)

- Outdoor IP camera: \$250
- Edge computer (Jetson Orin Nano/NX): \$350
- Outdoor display (high brightness): \$250
- Controller / Arduino (basic industrial-grade): \$80
- Case, power supply, cables, misc: \$120

👉 Total $\approx$ ¥7,200 CNY = ₩1,390,000 KRW = ¥148,000 JPY



Thank You

Q & A

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